

NSF Grant Proposal

Directorate for Biological Sciences - Division of Environmental Biology

Proposal Number: BIO-2024-07821
Program: Ecosystem Science
Program Officer: Dr. Katherine Wells
Submission Date: October 15, 2024

Project Information

Project Title: Alpine Ecosystem Response to Climate Change in the Northern Rockies
Principal Investigator: Dr. Sarah Chen
Co-PI: Dr. Michael Torres
Institution: University of Idaho
Department: Department of Biological Sciences
Requested Amount: \$485,000
Project Period: September 1, 2025 - August 31, 2028
Sponsoring Agency: National Science Foundation

Project Abstract

This project investigates the impacts of climate change on alpine ecosystems in the Northern Rocky Mountains. Our research combines long-term field monitoring, remote sensing, and ecological modeling to understand how rising temperatures, shifting precipitation patterns, and changing snowpack dynamics affect plant community composition, soil microbial diversity, and ecosystem carbon cycling. The study will establish 24 permanent monitoring plots across an elevational gradient (2,200-3,100m) in the Sawtooth and White Cloud mountain ranges.

Budget Overview

Category	Year 1	Year 2	Year 3	Total
Personnel	\$95,000	\$98,000	\$101,000	\$294,000
Equipment	\$35,000	\$5,000	\$5,000	\$45,000
Travel	\$12,000	\$14,000	\$14,000	\$40,000
Supplies	\$10,000	\$8,000	\$8,000	\$26,000
Indirect Costs	\$25,000	\$27,000	\$28,000	\$80,000
Total	\$177,000	\$152,000	\$156,000	\$485,000

Broader Impacts

This project will train 4 graduate students and 8 undergraduate researchers in alpine ecology and climate science. Results will be shared with the Sawtooth National Recreation Area and the U.S. Forest Service to inform land management decisions. We will develop educational materials for Idaho high school science teachers, reaching an estimated 500 students per year through a summer workshop program.